

A Weight-Free Benchmark for Multilateral Tariff Bargaining

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The question

Who gains how much from a trade agreement is what quantitative bargaining models were built to answer. The modern answer runs through *estimated bargaining weights*.

- **Bagwell, Staiger and Yurukoglu (2021)** — hereafter BSY — embed a Horn–Wolinsky network of bilateral Nash bargains inside a calibrated general-equilibrium trade model.
- They estimate, by GMM from the concessions countries actually made, the weight ζ that best rationalizes the Uruguay Round outcome.
- The estimated weights then carry the *division* of the gains.

The circularity

This is the right way to learn what bargaining power *was*. It cannot, by construction, say what the gains would have been divided into *absent* that bargaining power — the weights are recovered from the very outcome one would want to evaluate.

The literature has no weight-free yardstick against which an observed agreement can be read.

What a weight-free yardstick would have to do

Two requirements, and they pull against each other.

1. Contain no estimated weight

- The surplus a group can create, and the rule that divides it, must both be free of any parameter fit to the observed division.
- Otherwise the benchmark inherits exactly what it is meant to evaluate.

2. Live on the same economy

- Same countries, same calibrated primitives, same policy space, same threat point as the estimated model.
- Otherwise any gap between benchmark and bargain is a difference in setup, not in division.

The strategy of this paper

Hold BSY's calibration, panel and policy space fixed. Change *only* how surplus is generated and divided:

$$\underbrace{v(S)}_{\text{GE surplus a coalition can attain}} \longrightarrow \underbrace{\phi_i(v)}_{\text{Shapley division}}$$

Neither step contains a bargaining weight. The policy is chosen *before*, and independently of, the rule that divides its proceeds — so the benchmark is a *normative* object, not an estimate of power.

A cooperative benchmark, built inside BSY's own calibrated economy, and taken to their own panel.

1. **Replicate** BSY's economy — primitives, counterfactual and equilibrium map — and report honestly the one layer that does not replicate, their weight estimator.
2. **Value every coalition.** For each of the 63 coalitions S of the six negotiating countries, $v(S)$ is the general-equilibrium dollar surplus its members attain by choosing tariffs jointly, with non-members held at 1990 applied rates.
3. **Divide by axiom.** Allocate $v(N)$ by the Shapley value: efficiency, symmetry, dummy, additivity.
4. **Compare** to BSY's estimated bargained outcome — under explicit comparability conditions — and ask what the comparison does and does not support.
5. **Robustness:** repeat on the 31-country, 40-sector Caliendo–Parro (2015) economy, $N = 3$ to 7.

Agnew (2023, 2025) propose the Shapley value for trade gains in a stylized three-country example. We *derive* $v(S)$ from a structural GE model and enumerate all 63 coalitions exactly.

Four results, and what each survives

1. **The benchmark and the bargain disagree about Japan.** The estimated bargain awards Japan **55.5%** of the surplus; the benchmark assigns **42.9%**. Signs for US, Japan, Canada are robust across four allocation rules; **magnitudes are not** (US gap: +3.3 to +23.2pp), and a symmetric-weight placebo collapses the US gap to -0.06pp .
2. **The fair allocation is self-enforcing only with transfers.** The Shapley allocation is in the TU core (all 62 constraints, min slack 2.59%) — but without transfers the *efficient* policy is blocked by Australia alone.
3. **The shortfall is mostly agenda, not bargaining.** Of the 79.5% gap to a full-schedule optimum: coverage 62.2%, participation 1.2%, within-agenda 16.1%. Widening the agenda is worth $3.88\times$ bargaining harder.
4. **One external check — which fails its placebo.** Benchmark cuts track observed 1990–2000 liberalization ($\rho = +0.64$), but a model-free uniform cut scores higher ($\rho = +0.71$), resolved against us by paired bootstrap.

Plus a disclosed negative result: BSY's weight estimator does not reproduce, and we can say why.

Takes from

- *Quantitative Nash-in-Nash*: Bagwell–Staiger–Yurukoglu (2021), Collard–Wexler–Gowrisankaran–Lee (2019) — the economy and the comparator.
- *GE machinery*: Eaton–Kortum (2002), Caliendo–Parro (2015), Arkolakis–Costinot–Rodríguez-Clare (2012) — exact hat algebra.
- *Computation*: MPEC, Luo–Pang–Ralph (1996), Ossa (2014).
- *Reciprocity*: Bagwell–Staiger (1999, 2002) — Shapley’s marginal contribution is its formal counterpart.

Departs by

- Replacing the *estimated weight* with an *axiomatic allocation* — connects to cooperative treatments of trade agreements (Agnew 2023, 2025; Rogna 2016) and to the γ -core literature (Chander 2007; Ray–Vohra 2024).

Contribution

Make an axiomatic division *operational* inside a calibrated GE model — on the same panel and primitives an estimated bargaining model was fit to — and report what the comparison does and does not support.

The benchmark

Trade equilibrium: taken from the literature, unchanged

Multi-sector Eaton–Kortum à la Caliendo–Parro (2015), in exact hat algebra. N countries, J sectors, input–output linkages, Fréchet productivity, iceberg costs and ad-valorem tariffs.

$$\widehat{c}_n^j = \widehat{w}_n^{\gamma_n^j} \prod_k (\widehat{P}_n^k)^{\gamma_n^{k,j}}, \quad \widehat{P}_n^j = \left[\sum_i \pi_{ni}^j (\kappa_{ni}^j \widehat{c}_i^j)^{-\theta^j} \right]^{-1/\theta^j}, \quad \widehat{W}_n = \widehat{I}_n / \prod_j (\widehat{P}_n^j)^{\alpha_n^j},$$

plus gravity, expenditure and trade-balance conditions. Only base-year shares $\{\pi_{ni}^j, X_n^j\}$, elasticities θ^j and the tariff change enter.

Policy lever: one scalar per country

$$\tau_{ni}^{j,\text{new}} = s_n \tau_{ni}^{j,0}, \quad s_n \in [0, 1] \quad \forall n \in S, \quad s_n = 1 \quad \forall n \notin S.$$

- **Why:** cuts each coalition's decision space from $\sim N^2 J$ to $|S|$ — what makes exact enumeration over 2^N coalitions feasible. A tractability device, not a negotiating instrument (Results re-solves on BSY's own line-by-line instrument).
- $s_n \leq 1$ is **not a convenience**: it is the GATT bound-rate constraint, and it **binds** — four of six countries' unconstrained unilateral optimum is to *raise* tariffs.

The characteristic function

Value of a coalition

$$v(S) = \max_{s_S \in [0,1]^{|S|}} \sum_{n \in S} I_n^{\text{base}} (\widehat{W}_n(s_S, \mathbf{1}_{-S}) - 1), \quad v(\emptyset) = 0.$$

Three features, each a deliberate departure from the bargaining formulation it replaces:

1. **No bargaining weight anywhere.** Linear objective on a compact box $\Rightarrow$ a maximizer exists (Weierstrass) and any two maximizers attain the same value.
2. **No IR constraint inside the program.** “No member does worse than its outside option” is *relocated* to the allocation layer, where $\phi_n(v) \geq v(\{n\})$ is verified directly.
3. **Non-members are passive:** they keep 1990 tariffs and are not protected. This is plurilateral liberalization *with free-riding* — an MFN cut is a cut on outsiders too — not a discriminatory bloc.

$s_S = \mathbf{1}$ is feasible and yields the status quo, so $v(S) \geq 0$ constructively — not by assumption.

Why not a weighted Nash product inside $v(S)$?

The natural alternative fails on its own terms here.

- The Nash objective and the dollar payoff are *different functions*: the argmax of the product does not define the value of any single program.
- On coalitions whose individually rational set contains no strict Pareto improvement, the product is identically zero — and the numerical argmax reports the smallest dollar element of that face.
- It puts a bargaining weight back *inside* the characteristic function, which is exactly the object we are trying to build without one.

What is and is not weight-free — stated narrowly

The dollar-sum objective is a Negishi-type weighting of utilities by the inverse marginal utility of income: *not* distribution-neutral.

What *is* weight-free: the **division** is chosen after, and independently of, the policy — so alternative allocation rules operate on an identical $v(\cdot)$ and imply identical tariffs. (BSY themselves report a world-income-maximizing row; the objective is not novel here. [The allocation layer is.](#))

Cooperative normalization, and the single-step solve

Zero-normalization

$$w(S) \equiv v(S) - \sum_{n \in S} v(\{n\}), \quad w(\{n\}) = 0 \text{ by construction}, \quad \phi_n(v) = v(\{n\}) + \phi_n(w).$$

$v(\{n\})$ is what n gains *alone* — an outside-option effect on its own payoff, not cooperative surplus. We report $\phi(w)$ as the primary normative object and $\phi(v)$ when comparing to the estimated bargain: the comparator's gains are measured from the same 1990 status quo with nothing netted out, and **the two objects must share a threat point to be differenced**. On BSY's panel the distinction is small (four of six singletons are zero); on the second economy it is **decisive**.

Single-step formulation (MPEC)

$$\max_{\{s_n\}_{n \in S}, \widehat{w}, \widehat{P}, \widehat{c}, \widehat{\pi}, X'} \sum_{i \in S} l_i^{\text{base}}(\widehat{W}_i - 1) \quad \text{s.t. the equilibrium conditions and } s_n \in [0, 1].$$

Equilibrium variables are *decision* variables, not an inner loop: continuous quasi-Newton with exact implicit-function-theorem gradients, from multiple starts.

Grid or vertex enumeration would be invalid here — the grand-coalition optimum is interior on five of six coordinates and beats the best vertex by 17.4%.

Why the Shapley value

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (N - |S| - 1)!}{N!} (v(S \cup \{i\}) - v(S))$$

the unique rule satisfying **efficiency**, **symmetry**, the **dummy** property and **additivity**.

1. It nests the bilateral benchmark it generalizes.

Proposition (two players)

For $N = \{i, j\}$: $\phi_i(w) = \phi_j(w) = \frac{1}{2}w(N)$, the symmetric Nash bargaining solution with disagreement point $d_n = v(\{n\})$. On the raw game, $\phi_i(v) = v(\{i\}) + \frac{1}{2}w(N)$, which coincides with the symmetric Nash bargain measured from the 1990 status quo *iff* $v(\{i\}) = v(\{j\})$.

So it *extends* BSY's own bilateral benchmark to $N > 2$ rather than replacing it.

2. **It has the right economics.** Each country is paid its expected marginal contribution to a randomly formed coalition — the formal counterpart of the Bagwell–Staiger reciprocity intuition that leverage tracks the externality a country's tariffs impose on partners.
3. **It is computable.** Competing concepts are set-valued or lack closed form — fatal when the model must be re-solved at every one of 63 coalitions.

Comparing a normative object to an estimate

Fairness gap

$$\Gamma_i = \underbrace{\phi_i(v)/v(\mathcal{N})}_{\text{Shapley share}} - \underbrace{s_i^{HW}(\hat{\zeta})}_{\text{share of BSY's estimated bargained surplus}}$$

A difference of two shares is only meaningful if the two sides are defined on the same things. We state the conditions and check them.

- **Same threat point** — 1990 applied tariffs on both sides. (This is why the *raw* game v , not w , is the comparator's counterpart.)
- **Same player set** — all six negotiating countries, not the five that appear in bargaining pairs.
- **Same policy space** — cuts-only $s \in [0, 1]$; BSY's own solver caps every line at its 1990 rate.
- **Same coverage and instrument granularity** — our lever is one scalar per country over the *full* schedule; BSY's is 151 principal-supplier lines, set line by line. *These differ, so the level comparison is decomposed, not asserted (Result 3).*
- **Transferability regime** — BSY is NTU (their replication package hard-disables transfers); Shapley is TU. Disclosed, and carried as a separate term.
- **Normalization** — each side normalized on its own total; the difference between totals is carried explicitly as a level term, not hidden in the gap.

Data and replication

Data: BSY's 1990 negotiating panel, used unchanged

11 regions, 67 sectors, 1990. Six regions negotiate and are the players; the other five are “not elsewhere specified” composites, held fixed and never admitted as players (a composite is an accounting aggregate, not a strategic one). All 63 coalitions solved exactly.

Country	Applied tariff 1990 (%)	Income share (%)	Tariff revenue share (%)	Unilateral value $v(\{n\})$
United States	4.76	33.25	31.01	0.000
European Union	7.91	36.62	36.67	0.000
Japan	9.18	23.35	16.02	0.037
Canada	7.68	2.80	3.60	0.006
Australia	9.72	1.43	4.62	0.000
South Korea	12.26	2.55	8.08	0.000

Two features drive everything that follows:

- **Tariffs are low and fairly uniform** (4.8–12.3%), so the cuts-only lever is narrow and almost nobody gains from acting alone: $\sum_n v(\{n\})$ is only 6% of $v(\mathcal{N})$, and $\phi(v) = \phi(w)$ for four of six players.
- **Economic size is very unequal** — and here it is size, not tariff level, that drives marginal contributions.

Replication scorecard: three of four layers

Layer	Result	Verdict
Calibrated primitives (16 arrays, 3 scalars)	max abs. diff. 2.5×10^{-14} threshold 10^{-12} , pre-registered	reproduces
Published counterfactual	0.024 pp max abs. deviation	reproduces
Nash-in-Nash equilibrium map (cold start + box search)	0.40 pp mean tariff deviation but 26/151 lines on a different branch	reproduces with a caveat
GMM estimator of ζ	diverges: LM damping $7.4 \times 10^{12} \rightarrow 2.9 \times 10^{18}$	does not reproduce

Why the failure does not propagate

The characteristic function contains *no* bargaining weight, and every weight-dependent comparator is evaluated at BSY's own published $\hat{\zeta}$ and their own cached equilibrium. It does, however, motivate the placebo and branch checks later: a comparator recovered from a fragile estimation problem deserves them.

The estimator failure is a result, not an accident

The mechanism is identifiable, and it is a statement about the *estimated model*, not about our implementation.

1. Each bilateral pair in the Nash-in-Nash fixed point reaches a deal on a line *only if* its minimum individual gain clears a tolerance — and reverts to the 1990 baseline otherwise.
2. So the equilibrium is a **step function of the weight vector**, with a discrete deal/no-deal branch on every line.
3. **Direct evidence:** two independent re-solves of the *identical* symmetric point $\zeta = 0.5$ return criterion values differing by 4.1×10^{-4} ; an isolated, fully converged re-run of that same point lands on a branch 5.08 pp away from the cached comparator.
4. The finite-difference step used to assemble the Jacobian — chosen to match BSY's own minimum step — is **of the same order as this branch noise**.

Reading

At that resolution the GMM criterion is better described as a *path-dependent step function* than as a smooth objective — also the most likely reason BSY's own estimation required a coarse minimum differencing step.

Results

Result 1: the benchmark allocation

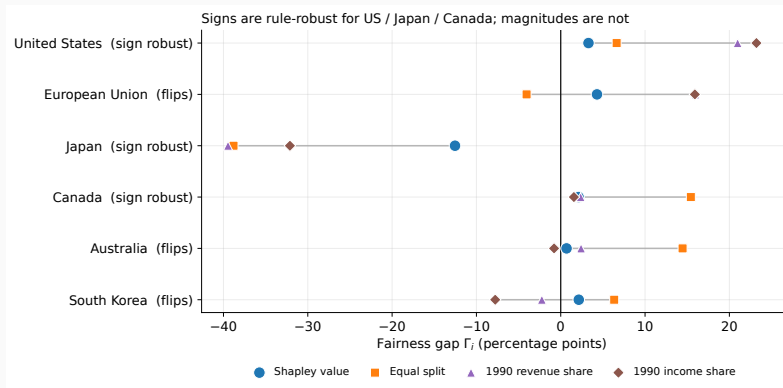
$v(\mathcal{N}) = 0.720$ in BSY's income units, of which 0.677 (94.05%) is genuinely cooperative surplus. Optimal policy $s^* = (0.53, 0.47, 0.03, 0.33, 0.11, 0.00)$: interior, strongly asymmetric, and 17.4% above the best corner of the box.

Country	Shapley benchmark $\phi_i(v)$	share (%)	Bargained share (%)	Efficient policy incidence (%)	Fairness gap (pp)	Transfer T_i
United States	0.096	13.33	10.04	15.82	+3.29	-0.018
European Union	0.180	25.02	20.71	29.15	+4.30	-0.030
Japan	0.309	42.93	55.47	49.82	-12.54	-0.050
South Korea	0.090	12.46	10.33	6.46	+2.14	+0.043
Canada	0.024	3.36	1.24	2.84	+2.12	+0.004
Australia	0.021	2.90	2.21	-4.08	+0.69	+0.050

$\Gamma_i = \phi_i(v)/v(\mathcal{N}) - s_i^{HW}(\hat{\zeta})$; bargained shares are of BSY's cached equilibrium total $\sum_k g_k^{HW} = 0.3124$. Incidence is $g_i(s^*)/v(\mathcal{N})$, what each country gets at the surplus-maximizing policy *before* any reallocation; $T_i = \phi_i(v) - g_i(s^*)$, positive means the country receives.

The two allocations disagree about Japan — in one direction and by a wide margin. Every other member is under-compensated relative to the benchmark.

The gap is robust in sign, not in magnitude



Four *efficient and individually rational* rules on the same $v(\cdot)$, comparator $s^{HW}(\hat{\zeta})$ held fixed: **sign-robust** for US, Japan, Canada; **signs flip** for EU, Australia, South Korea; and **magnitudes are strongly rule-dependent** — the US gap ranges +3.29 to +23.21pp, a factor of seven.

Direction for US/Japan/Canada is rule-independent. Any statement about *size* is a statement about the Shapley value specifically — so we never quote a magnitude without naming the rule.

What the gap actually measures

Neither $\hat{\zeta}$ nor either equilibrium branch is privileged by the model. So vary the *comparator* side — the benchmark side $\phi_i(v)/v(\mathcal{N})$ does not depend on ζ at all.

Γ_i (pp)	US	EU	Japan	Canada	Australia	S. Korea
At $\hat{\zeta}$, cached branch (primary)	+3.29	+4.30	−12.54	+2.12	+0.69	+2.14
At $\hat{\zeta}$, alternative branch	+5.81	+10.99	−13.78	+3.04	−0.91	−5.15
At $\zeta = 0.5$, symmetric placebo	−0.06	−0.76	+6.79	+1.97	+0.47	−8.41

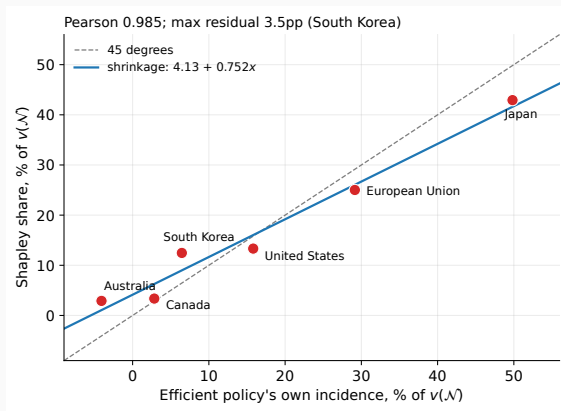
The placebo is the discipline

At a symmetric weight the Shapley benchmark and a Horn–Wolinsky bargain award the United States *almost exactly the same share* ($\Gamma_{US} : +3.29 \rightarrow -0.06\text{pp}$), and four of six signs flip.

⇒ The US gap at $\hat{\zeta}$ is a readout of **how asymmetric BSY's estimated weights are**, not evidence of a structural distance between cooperative benchmarks and bargaining outcomes.

The sign pattern we *do* report — US and Canada positive, Japan negative — holds on both equilibrium branches. That is why we report signs.

Most of the benchmark is a rescaling of physical incidence



97% of the cross-country variation in the benchmark is already in the efficient policy's *own* incidence. A one-parameter shrinkage — slope 0.752, intercept 4.13pp — reproduces the Shapley share to within 3.5pp at every player.

In its favour. What the axioms add is one describable operation: a compression of dispersion toward players with positive marginal contribution but small or negative incidence — precisely what symmetry and the dummy property are *for*. Australia and South Korea are the two players it protects.

Against a strong reading. A referee reproduces 97% of the pattern with the shrinkage rule alone, and the largest place Shapley adds information beyond it (3.5pp, South Korea) is the same order as the headline US gap itself.

n = 6, two-parameter fit: no significance statement is available, and we make none.

Result 2: stability must be checked, not inferred

The usual route is closed. The game is not convex — and not even superadditive.

What fails

- 20 of 301 disjoint-pair superadditivity tests fail.
- Single-player monotonicity of w fails in 17 of 192 additions, worst margin -0.026 .
- Failures are **not** high-order: they start at the *pairwise* level. $\{\text{US}, \text{Canada}\}$ is worth less cooperatively than the two apart — Canada's unilaterally profitable cut degrades US terms of trade by more than it gains Canada.
- Worst case: US joins $\{\text{EU}, \text{Canada}\}$ and value falls $0.030 \rightarrow 0.004$ ($w: +0.024 \rightarrow -0.002$). The EU captures the gains from Canada's opening; the US, once inside, bears its cost.
- **Canada is in 16 of the 17 violations.**

What we verify directly

- **Individual rationality:** $\phi_i(v) \geq v(\{i\})$ at every player. Minimum slack $0.0186 = 2.59\%$ of $v(\mathcal{N})$ (Canada).
- **Core membership:** all 62 proper-coalition constraints satisfied, same binding coalition, same slack.

Why it survives

Shapley averages marginal contributions over orderings, and that averaging absorbs a local concavity whose deepest point is 3.7% of $v(\mathcal{N})$ and which is confined to 17 of 192 additions.

On this economy an argument for IR or core membership that runs through monotonicity or convexity is simply wrong. The conclusions survive only because they were checked.

Core membership is conditional in two ways that matter

(1) Passive outsiders (A4)

All 62 constraints hold non-members at their 1990 tariffs. That convention is not innocuous here — the US's outsider gain is *negative* for 13 of the 31 coalitions excluding it — and we have **not** shown the result survives a γ -response convention in which outsiders best-respond (Chander 2007). [The single largest robustness exercise we leave open.](#)

(2) The verified core is the core of the *transferable-utility* game

The one transfer-free object the model produces — the efficient policy's own incidence — is **not in it**: Australia's incidence is -4.08% , so **Australia alone blocks the surplus-maximizing policy**.

Moving from that incidence to the benchmark needs a budget-balanced gross transfer of **13.5% of $v(N)$** : the US, EU and Japan pay 2.5, 4.1 and 6.9%; Australia receives 7.0%.

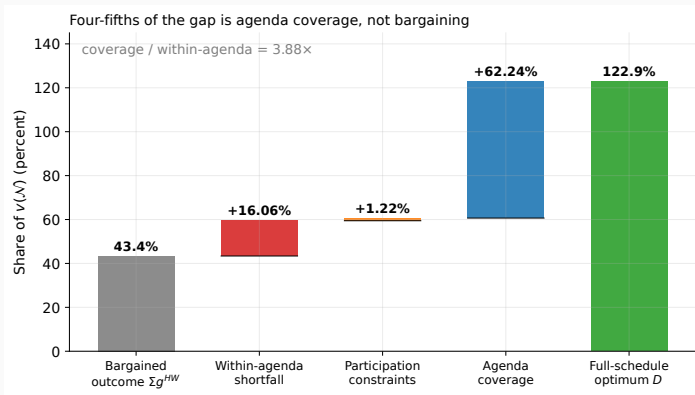
The GATT/WTO is by design a no-transfer institution

It operates through issue linkage and reciprocal concessions, not side payments. So the correct statement is conditional in both directions:

the fair allocation is self-enforcing *if* transfers are feasible — and if they are not, even the efficient policy is blocked by a single member.

Result 3: what the bargain left on the table

$\sum_k g_k^{HW} = 0.312$ against $v(\mathcal{N}) = 0.720$ — a level gap of 57% that *invites*, and does not support, the reading that bilateral bargaining left half the gains unrealized. Three things vary at once: coverage, granularity, transferability regime.



Re-solved on BSY's own line-by-line instrument — their 151 negotiated lines, then the full 294-line schedule. Each term varies exactly one dimension.

Three readings follow — and one does not

1. **The shortfall is mostly about the agenda.** Coverage is worth 62.2% of $v(\mathcal{N})$ against 16.1% for the within-agenda residual; on the regime-matched comparison the ratio is **3.88 to one**. Widening what is negotiable is worth about four times more than bargaining harder over what already is. And the excluded set is not arbitrary — it is BSY's own estimation sample, built from reciprocal principal-supplier relationships; agriculture leaves it by construction.
2. **Transferability is not the story.** The IR constraint costs 1.2% of $v(\mathcal{N})$ — 2.0% of the unconstrained objective.
3. **The residual is not an artifact of the 1990 threat point.** Replace 1990 with BSY's own realized outcome as every member's floor and an attainable point worth 0.4109 remains, 13.7% of $v(\mathcal{N})$ above the bargained outcome — preserving 85% of the within-agenda shortfall.

What does *not* follow

Attributing the residual 16.1% to bargaining *skill*. Two things stay jointly embedded in it: how deep the agreed cuts were on shared lines, and what a seven-pair bilateral architecture can deliver relative to a grand-coalition optimizer. We tried to separate them and got a **boundary result, not a verdict** (backup).

The improvement is real, weak, and unevenly shared

Anchoring participation at what negotiators actually delivered

Replace the 1990 threat point with BSY's *own realized outcome* as every member's floor. An attainable point worth $F^{PD} = 0.4109$ survives: 13.7% of $v(\mathcal{N})$ above the bargained outcome, and no-transfer-feasible.

The qualifier is the finding

It is a **weak** Pareto improvement. Four of the six members sit *exactly* at their own floor, and the entire net gain accrues to Japan and Canada.

Two auxiliary checks support the within-agenda reading:

- BSY's cached equilibrium moves *no* tariff line outside the 151-line mask — 0 of 251 off-mask cells — so “within-agenda” is a *measured* property of their solution, not an assumption.
- Their cached schedule is very nearly, but not exactly, feasible in our cuts-only box: 5 of 151 lines (3.3%) lie below zero — legal under their lower bound of -0.25 , outside our floor of zero.

So the three decomposition terms are non-negative as a fact about these computed numbers, verified term by term — not as a geometric guarantee.

Result 4: the one check against data outside the model

Everything so far is model-internal. The one external comparison available: the benchmark's own optimal line-by-line cuts against the *observed* 1990–2000 change in applied MFN tariffs on the identical 151 lines.

Cut schedule	Spearman ρ	Lines cut
Benchmark, IR-constrained (F^{IR})	+0.641	118 of 151
Benchmark, unconstrained (F^{v2})	+0.667	—
BSY's own fitted schedule at $\hat{\zeta}$	+0.317	65 of 151
Uniform-cut placebo (model-free)	+0.714	151 of 151

It clears the pre-registered band — and fails its placebo

A rule with *no model in it at all* — cut every line by the same proportion, i.e. rank lines by their 1990 tariff — scores higher, and a paired bootstrap resolves the difference **against** the benchmark ($\Delta\rho = -0.073$, 95% CI $[-0.148, -0.005]$).

Honest statement: the benchmark's cuts point the same way as observed liberalization but **carry no information about it beyond persistence of the 1990 tariff**. Same for BSY's fitted schedule — once rank is partialled on the 1990 tariff, $|\rho_{\text{partial}}| < 0.05$ for both.

Head-to-head with BSY: sharper on the extensive margin

- BSY's cached equilibrium leaves 86 of 151 lines (57%) at their solver's no-cut tolerance — on lines the data show were, on net, cut over the decade. The benchmark cuts 118.
- Pooled rank comparison favours the benchmark and a paired bootstrap resolves it: $\Delta\rho = +0.324$, CI [+0.204, +0.459].
- But the two schedules do not have the same coverage — a schedule cannot lose rank ties on lines it never moves. On the 65 lines where *both* move: $\Delta\rho = +0.061$, CI [−0.014, +0.147] — not resolved.

We take the coverage-matched comparison as terminal, and report the head-to-head as unresolved.

Two checks explain the gap between the two calipers, and neither gives a simple story:

- Pinning a *random* 86 of the benchmark's lines at BSY's no-cut value, 5,000 times, gives mean $\rho = +0.132$; BSY's actual +0.317 sits at the 98.7th percentile. BSY's choice of which lines to leave uncut is itself informative — the tie count alone does not explain their lower score.
- The benchmark's own cut pattern resembles the model-free placebo far more than BSY's fitted schedule does ($\rho = 0.88$ vs 0.50), so part of the benchmark's pooled advantage is its own greater resemblance to naive tariff persistence, not superior structural content.

Registered predictions that failed

We registered these tests before running them. Four registered claims failed, and they are in the body of the paper, not only the appendix.

1. **US fairness gap** was registered to clear +5pp. It is +3.29pp on the primary branch — it clears only on the alternative branch.
2. **US passive outsider gain** was registered to be negative on a *majority* of coalitions excluding it. It is negative on 13 of 31.
3. **Grand-coalition optimum** was registered to be interior at every player. South Korea's optimum is exactly at the boundary; only the weakened form — five of six interior — holds.
4. **Retracted claim.** An earlier version claimed implementing the benchmark requires a transfer of about 19% of $v(N)$ to the United States, on the premise that the US loses at the efficient policy. **That is retracted:** at the certified optimum the loser is Australia, and the US *pays* into the reallocation.

The external check (previous slide) is a fifth: it cleared its registered band and then failed its own placebo. We report it that way.

Robustness and scope

A second economy: Caliendo–Parro (2015), $N = 3$ to 7

BSY's panel is six similar, low-tariff economies. The CP 1993 economy is heterogeneous in exactly the dimension BSY's is not: applied tariffs 3.4% to 34.5%. Coalitions enumerated exactly from NAFTA ($N = 3$) to NAFTA + China, India, Brazil, Korea ($N = 7$).

What transfers

- Core membership: all 126 constraints hold at $N = 7$, min slack 0.37% of $v(N)$. IR at every N .

What does *not* — informatively

- The *failure* of the sufficient conditions. w -monotonicity, violated 17/192 on BSY's panel, holds on all 3,019 tested additions here; supermodularity fails on 0.61% of triples.
- A game from the same primitives class can be well behaved or not depending on the panel. Neither can be assumed.

Substantive contrast: composition of surplus

- BSY: 94.05% of $v(N)$ is cooperative. CP with a high-tariff member ($N \geq 4$): only 13.1–16.1%, against 75.2% at $N = 3$. The rest is unilateral gain (China \$119.7bn, India \$26.4bn).
- Raw $\phi(v)$ tilts to China and India; **cooperative $\phi(w)$ reverses it** — the top cooperative contributor at every $N \geq 4$ is the **US** (35.5–45.2% of $w(N)$).

Adverse horse race

Raw shares are *not* a relabeled GDP share (Pearson -0.28 to $+0.07$) — but at $N \geq 4$ they are near-indistinguishable from a relabeled **tariff-revenue** share (0.84–0.87). When one member's headroom dominates, marginal contributions track headroom.

What we do not claim

The allocation rule is not identified — and the reason is sharper than the generic caveat

Following Manski (1995), identification requires that distinct parameter values generate distinct distributions of observed outcomes.

Here the policy that generates $v(S)$ is chosen *independently* of the rule that divides the surplus. So the Shapley value, the nucleolus, Kalai–Smorodinsky and Chander's γ -core **all imply exactly the same tariffs at every coalition**. No tariff moment can discriminate among them — only observed international transfers could, and the GATT does not generate them.

Therefore we do not claim to: identify bargaining power; identify the allocation rule; or read the Shapley ranking as evidence about strategic position.

Structural limits, not incidental ones:

- **Threat point.** The 1990 applied schedule is itself the outcome of four decades of prior GATT bargaining — “fairness relative to the status quo” inherits whatever the status quo encodes.
- **Passive outsiders.** A γ -response convention would change every proper-coalition value and hence the stability results. Left open.
- **Of our own making:** one scalar per country in the benchmark itself, and no data moment on any coalition value other than the grand coalition.

Takeaways

1. **Raw cooperative shares are not leverage rankings.** With heterogeneous protection, most of a high-tariff country's raw share is what it could capture *unilaterally*; netting it out reverses the ranking and puts the large, import-diversified economy on top. A GDP- or volume-based weighting would miss this — and so would reading the raw ranking on its own terms.
2. **A fair allocation and an implementable one are different objects**, and the gap is first-order here. Core membership says the benchmark is self-enforcing *when transfers are feasible*. In a no-transfer institution the efficient policy is not merely unattractive to Australia — it is *blocked* by Australia. Any benchmark of this type owes its reader the transfer-free counterpart.
3. **Decompose an efficiency shortfall before attributing it.** Comparing a full-schedule planner to a bilateral bargain crosses coverage, granularity and participation at once. Held fixed, the residual attributable to bargaining performance is a *fifth* of the raw gap — and even that still mixes cut depth with the bilateral architecture.

Two extensions worth the cost

An NTU / λ -transfer version of the benchmark, which would close the one comparability gap no convention can; and the γ -response recomputation, which would tell us whether the core result is a property of this economy or of the passivity convention. Both are computationally heavy rather than conceptually hard: same program, different constraints.

**What an agreement would have looked like
had it divided its gains by an axiom
rather than by power.**

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Backup: the MPEC in full

$$\begin{aligned}
 & \max_{\{s_n\}_{n \in S}, \widehat{w}, \widehat{P}, \widehat{c}, \widehat{\pi}, X'} \sum_{i \in S} l_i^{\text{base}} (\widehat{W}_i - 1) \\
 & \text{s.t.} \quad \widehat{c}_n^j = \widehat{w}_n^{\gamma_n^j} \prod_k (\widehat{P}_n^k)^{\gamma_n^{k,j}}, \\
 & \quad \widehat{P}_n^j = [\sum_i \pi_{ni}^j (\widehat{\kappa}_{ni}^j \widehat{c}_i^j)^{-\theta^j}]^{-1/\theta^j}, \quad \widehat{\pi}_{ni}^j = (\widehat{c}_i^j \widehat{\kappa}_{ni}^j / \widehat{P}_n^j)^{-\theta^j}, \\
 & \quad X_n^{j'} = \sum_k \gamma_n^{j,k} \sum_i \frac{\pi_{in}^{k'} X_i^{k'}}{1 + \tau_{in}^{k'}} + \alpha_n^j l_n', \\
 & \quad \sum_{j,i} \frac{\pi_{ni}^{j'} X_n^{j'}}{1 + \tau_{ni}^{j'}} - D_n = \sum_{j,i} \frac{\pi_{in}^{j'} X_i^{j'}}{1 + \tau_{in}^{j'}}, \\
 & \quad \widehat{\kappa}_{ni}^j(s_n) = (1 + s_n \tau_{ni}^{j,0}) / (1 + \tau_{ni}^{j,0}), \\
 & \quad s_n \in [0, 1] \quad \forall n \in S, \quad s_n = 1 \quad \forall n \notin S.
 \end{aligned}$$

One trade-balance constraint is redundant by Walras' law and is dropped. Deficits D_n are set to zero in every counterfactual, so welfare changes are not driven by an exogenous transfer. Solved by continuous quasi-Newton optimization with exact implicit-function-theorem gradients through the equilibrium, from multiple starts.

Backup: fairness gap under four allocation rules

Rule	US	EU	Japan	Canada	Australia	S. Korea
Shapley value	+3.29	+4.30	-12.54	+2.12	+0.69	+2.14
Equal split	+6.63	-4.05	-38.81	+15.43	+14.45	+6.34
1990 revenue share	+20.98	+15.95	-39.46	+2.36	+2.41	-2.24
1990 income share	+23.21	+15.91	-32.12	+1.56	-0.79	-7.77
Sign-robust	yes	no	yes	yes	no	no

All four rules are **efficient and individually rational** on $v(\cdot)$, and the comparator $s^{HW}(\hat{\zeta})$ is held fixed — so this table varies the *allocation rule* only.

It is a different question from the ζ -placebo, which varies the *comparator* holding the rule fixed. The two checks are not substitutes.

Computed in-sample from the same coalition values as the headline table: this is a sensitivity check, not an out-of-sample test. With $n = 6$ no significance statement is available.

Backup: is there a weight that reproduces the benchmark?

The question

Does any weight vector ζ^* in BSY's own box reproduce the Shapley shares? If one existed, the benchmark could be read as an implicit bargaining-power statement.

What we did. Searched from eight starting points, including $\hat{\zeta}$ and the symmetric point.

We report no answer — not a null result

Every start exhausted its compute budget **assembling its first Jacobian**: a single equilibrium evaluation takes on the order of fifteen minutes and seven perturbations are needed per step. The objective's differentiability was never reached.

And the same branch-flip mechanism that defeats the weight estimator is what makes the assembly expensive in the first place.

8/8 starts stall on *wall-clock*, not on the objective. Reporting this as “no such weight exists” would be a category error, so we do not.

Backup: the decomposition in numbers

	Instrument, coverage, regime	Value	Share of $v(\mathcal{N})$
$\sum_k g_k^{HW}$	BSY's bargained outcome, 151 lines	0.3124	43.4%
F^{IR}	Line-by-line, 151 lines, IR-constrained	0.4281	59.4%
F^{v2}	Line-by-line, 151 lines, unconstrained	0.4368	60.7%
D	Line-by-line, 294 lines, unconstrained	0.8850	122.9%
<i>Decomposition of $D - \sum_k g_k^{HW} = 0.5726$</i>			79.51%
Agenda coverage	$D - F^{v2}$, coverage varies only	0.4482	62.24%
Participation constraints	$F^{v2} - F^{IR}$, regime varies only	0.0088	1.22%
Within-agenda shortfall	$F^{IR} - \sum_k g_k^{HW}$	0.1156	16.06%

F^{IR} imposes each member's own line-by-line unilateral value as its floor; a uniform zero floor instead gives 0.4285 and a within-agenda term of 16.12% — an *upper bound*, superseded. Components sum to 79.52% by independent rounding; the exact total is 79.51%.

D exceeds $v(\mathcal{N})$ because a line-by-line instrument is strictly finer than one scalar per country. That difference — 22.9% of $v(\mathcal{N})$ — is the **instrument-granularity dividend** at full coverage and is *not* part of the decomposition. It is coverage-dependent, so it is never quoted without the coverage level at which it was measured.

Backup: the attempt to separate cut depth from architecture

What we tried

Re-solve BSY's *own* pair structure under our objective: each of the seven principal-supplier pairs maximizes its own two members' combined welfare, iterated to a fixed point.

Boundary result, not a verdict

The resulting value is 0.3114 — it falls *just below* the bargained outcome it was meant to bracket (0.3124), by 0.15% of $v(\mathcal{N})$. That triggers the pre-registered boundary case rather than either conclusion. We report it as a diagnosis, not a decomposition.

Two by-products that do not depend on the failed separation:

- Replacing a joint 151-line planner with the seven-pair architecture costs 17.4% of $v(\mathcal{N})$ on a transfer-unconstrained basis.
- About **a quarter (26%)** of the surplus any one pair creates for itself spills, uninternalized, to the four countries outside it — structurally analogous to BSY's own finding that free-rider gains account for 35.4% of the Round's realized world-welfare gain.

A caveat we disclose: a within-pair dollar-utilitarian planner is a different solution concept from BSY's within-pair Nash product with an estimated weight and each pair's own 1990 disagreement point — not two readings of the same object.

Backup: numerical certification, and what it licenses

Multi-start certifies two logically distinct things — we separate them

- **Value certification.** For every object in the decomposition, the maximized value agrees across starts to at least 10^{-9} .
- **Argmax certification.** Does *not* always hold. The full-coverage optima D and D^{IR} sit on a nearly flat ridge — per-line differences across starts up to 0.15 — while the 151-line optima F^{v2} , F^{IR} and the Pareto-dominant point agree line by line to 6×10^{-7} or better.

Consequence, enforced throughout: D and D^{IR} are quoted *only as values* — they enter the decomposition and nothing else — and every line-level claim, including the rank comparisons against observed cuts, is confined to objects whose argmax is pinned down.

Solve noise is smaller still:

- Fixed-policy GE re-solve dispersion: 7×10^{-13} relative.
- Optimizer-restart dispersion: order 10^{-12} .
- Multi-damping agreement 2.5×10^{-5} ; 127 of 127 bitwise identical re-solves.
- Identity counterfactual at $s = \mathbf{1}$: $\max_i |g_i| = 2.2 \times 10^{-13}$.

Numerical error contributes nothing at the precision we report. The uncertainty that matters is rule dependence and branch sensitivity — both orders of magnitude larger.

Backup: second-economy numbers

Object	Value
Unilateral values $v(\{n\})$	China \$119.7bn, India \$26.4bn, Mexico \$1.9bn, Brazil \$1.7bn; US / Canada / Korea exactly 0
Cooperative share $w(N)/v(N)$	13.1–16.1% at $N \geq 4$; 75.2% at $N = 3$
Top cooperative contributor	United States, 35.5–45.2% of $w(N)$ at every $N \geq 4$
Core, $N = 7$	0/126 violations, min slack 0.37% of $v(N)$
w -monotonicity	3,019/3,019 pooled; 2,059/2,059 at $N = 7$
Supermodularity	fails 31/5,103 (0.61%) at $N = 7$
Horse race vs GDP share	Pearson -0.28 to $+0.07$
Horse race vs tariff revenue	Pearson 0.84 – 0.87 at $N \geq 4$; $+0.29$ at $N = 3$

Why the tariff-revenue horse race is reported even though it is adverse. It bears directly on how much discriminating work the Shapley structure does on an economy of this shape: when one member's headroom dominates, marginal contributions track headroom.

On BSY's own panel, where headroom is uniform, the same horse race is inconclusive at $n = 6$ and its band placement is driven entirely by Japan — dropping Japan moves every leg into the adverse band.

Neither panel supports a claim that the Shapley allocation does something a simple tariff-revenue weighting could not.